

INTERNSHIP REPORT

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Udaya Sri



Department of Electronics And Communication

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**JNTUH UNIVERSITY COLLEGE OF ENGINEERING
SULTANPUR**

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ABSTRACT

Agriculture is an integral part of Indian economy. Over 60% of Indian population based upon agriculture and one third of the income of nation arises from agricultural practices. Hence it plays a vital role in the development of the country. Various issues related to farming is continuously hampering the development of the country. Possible solution for these problems is to opt for modernized agriculture that comprises of modern trends. Hence, agriculture can be made smart using IoT and other technologies. Smart agriculture increases crop yield, decreases water wastage and imbalanced use of fertilizers.

The highlighting feature of this project is that it measures the different agricultural parameters affecting the yield and it also uses a GPS module to get the information about the location. Secondly it sends all the data to the cloud where it can be further analyzed.

Thirdly this project also contains an android mobile app providing an easy access of information to the farmer. Moreover this project presents a smart irrigation system that optimizes water usage.

CHAPTER 1

INTRODUCTION

1.1 Introduction

Internet of Things (IoT) is the interconnection or network of physical devices that is interrelated computing devices, digital and mechanical machines, people or animals, objects that can sense, accumulate and transfer data over web without any human involvement. Everything is provided with unique identifier. It is a progressed examination and mechanized frameworks which uses detecting, organizing, enormous information and man-made consciousness innovation to convey total framework for an administration. Basically IoT is about extending the power of internet beyond smart phones and computers.

IoT has changed today's world. Smart cities, smart car, smart homes everything around us can be turned into a smart device with the help of IoT. It also has applications in agriculture, business sectors, healthcare, transport and logistics. There are four main components of IoT-

- Low power embedded system- High performance and less battery consumption are the inverse factors that play an important role in design of electronic system.
- Cloud computing- Data collected from devices is stored on reliable storage servers so here cloud computing comes into action.
- Availability of Big Data- As IoT is highly dependent on sensors that are real time. So the usage of electronic devices is spread throughout every field that is going to trigger a massive flux of data.
- Network connection- For communication, internet connectivity is necessary where each physical object is assigned by an IP address. A network connection is build between the devices with the help of these addresses 1 Technology today has not reached its 100% capability. So the advantages and disadvantages of this technology are given below-

Advantages of IoT

1. Utilization of Resources Efficiently
2. Minimization of Human Efforts
3. Time-saving
4. Increase Data Collection

Disadvantages of IoT

1. Security
2. Privacy
3. Complexity

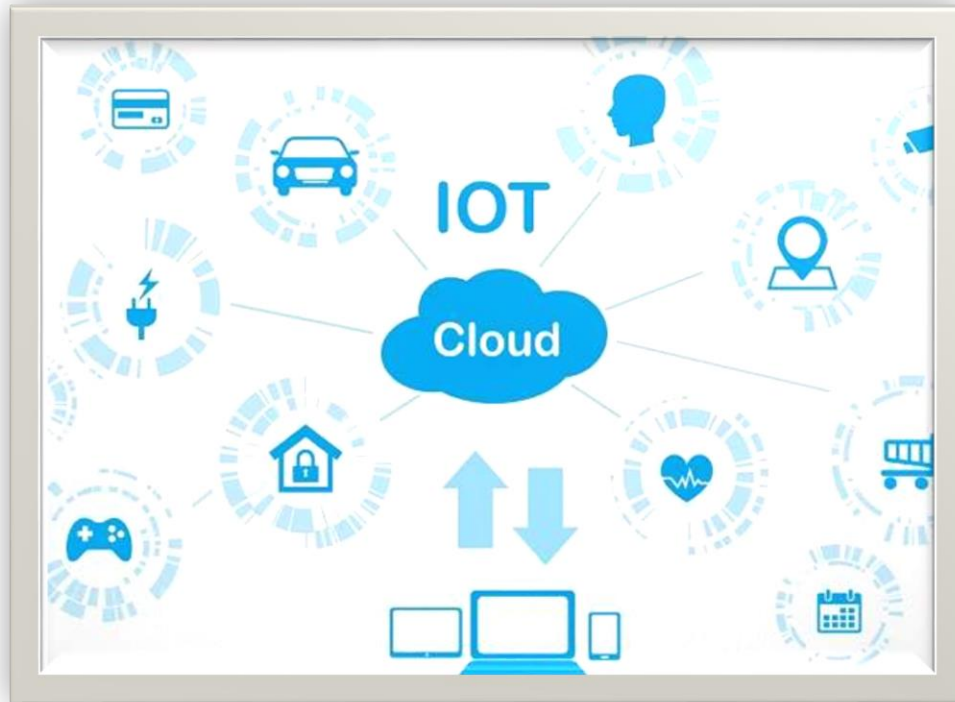


Figure 1.1 Description of IoT

1.2 IoT in Agriculture

Internet of Things has capacity to transform the lives of people in the world in an efficient manner. The ever growing population would touch more than 3 billions in few years. So to feed such an immense population, agriculture industry need to embrace IoT. The demand for more food has to address challenges that include excessive climate conditions, weather change and different environmental affects that results from farming practices. The destiny of Indian agriculture must be worked with understanding and excessive cease technologies that can expand production and furthermore regains the attention of farmers in this industry. So these smart farming techniques would assist farmers to lessen scrap and enhance capacity. It is basically a high tech and capital intensive system for growing crops in a sustainable manner for masses. This technology can help farmers to monitor field conditions from anywhere with the help of sensors and can also irrigate fields with an automated system. It is the application of Information and Communication Technology into the field of agriculture.

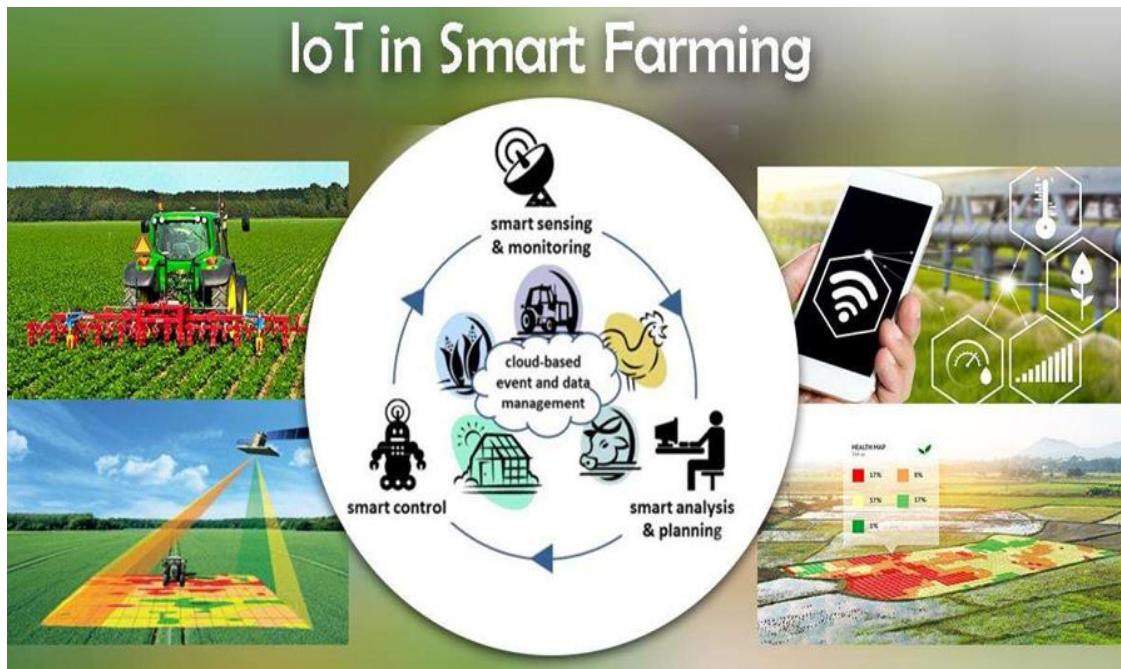


Figure 1.2 Smart Farming

1.3 Structure of IoT in Agriculture

Basically this system structure consists of 3 layers that are sensor layer, transport layer, application layer and the functions of these layers are below –

1. Sensor layer- One of the challenge of the sensor layer is to obtain automated and real time transformations of the figures of actual world agricultural manufacturing into digital transformation or information which could be processed in virtual world through different or various means. The data that they collect are-
 - Sensor information- Humidity, temperature, gas concentrations, pressure etc.
 - Products information- name, model, price and features.
 - Working condition - operating parameters of different equipments, apparatus etc.
 - Location information

The major challenge of Information layer is to mark diverse kinds of information or data and gathering the information and marked information in the actual world by means of techniques of sensing, after which remodels them for processing into digital information. This sensor layer includes some strategies- RFID tags, cameras, two dimension code labels, sensor networks.

- 2) Transport layer- This layer's task is to acquire and summarize the data of agriculture acquired from the above layer for processing. It is believed as the nerve centre of IoT. This layer includes the combination of telecommunication management centre and also internet network, information centre, smart processing centres.

3) Application layer- The function of this layer is to analyse and process the information collected for the cultivation of digital awareness of actual world. It is considered as a fusion of IoT and agricultural market intelligence.

Benefits of IoT in Agriculture

- IoT empowers simple gathering and the executives of huge amounts of information which is gathered from sensors used and with the help of joining of distributed evaluating administrations such as cloud storage, farming field maps and more information can be retrieved from any place and everywhere which enables live monitoring and connectivity which is end to end.
- IoT is viewed as an important segment for smart farming because with precise use of sensors and also the smart gadgets, farmers could expand the output by 72% upto year 2050 as delineated by specialists.
- By the use of IoT creations expenses could be diminished to an astounding dimension that would thus expand productivity and survivability.
- By the use of IoT efficiency level would be further expanded as far as utilization of water, soil, fertilizers, pesticides etc.

Applications of IoT in Agriculture

- Precision farming
- Agricultural drones
- Livestock monitoring
- Smart greenhouses

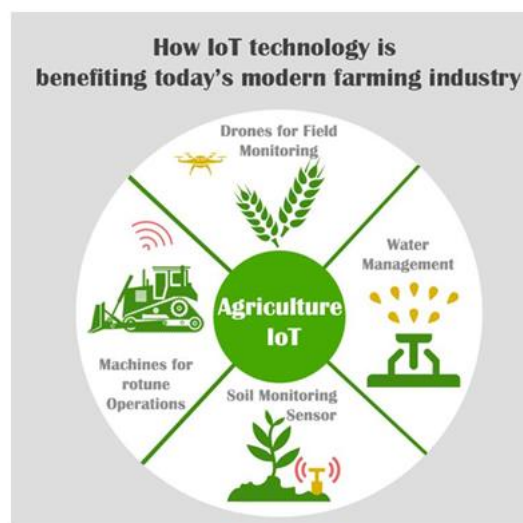


Figure 1.3 Applications of IoT

CHAPTER 2

SYSTEM DEVELOPMENT

2.1 Methodology

The basic building blocks of an IoT System are Sensors, Processors and applications. So the block diagram below is the proposed model of our project which shows the interconnection of these blocks. The sensors are interfaced with Microcontroller, data from the sensor is displayed on the mobile app of the user. Mobile app provides an access to the continuous data from sensors and accordingly helps farmer to take action to fulfil the requirements of the soil.

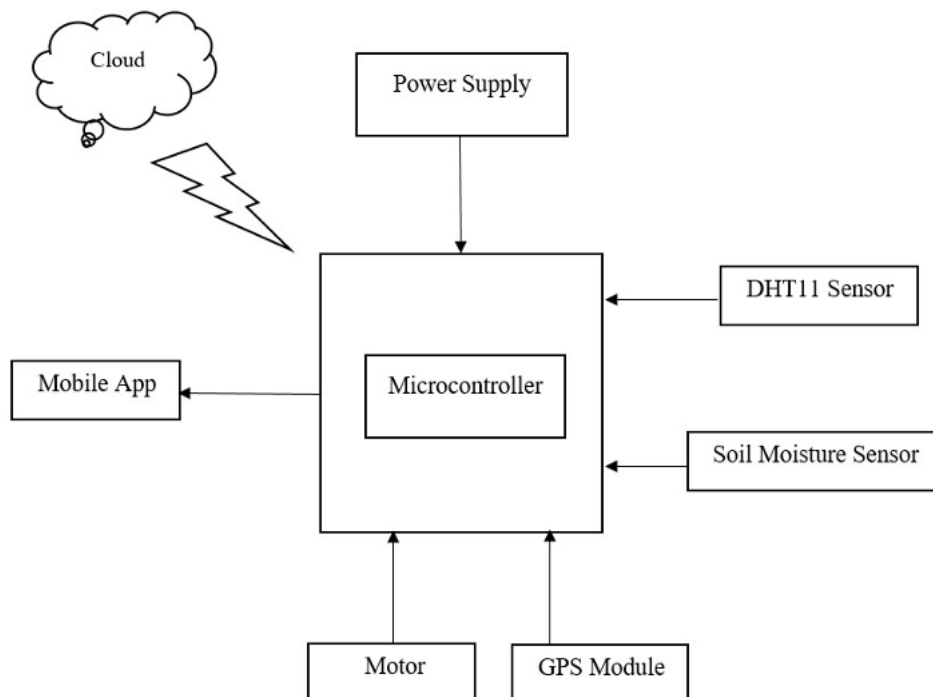
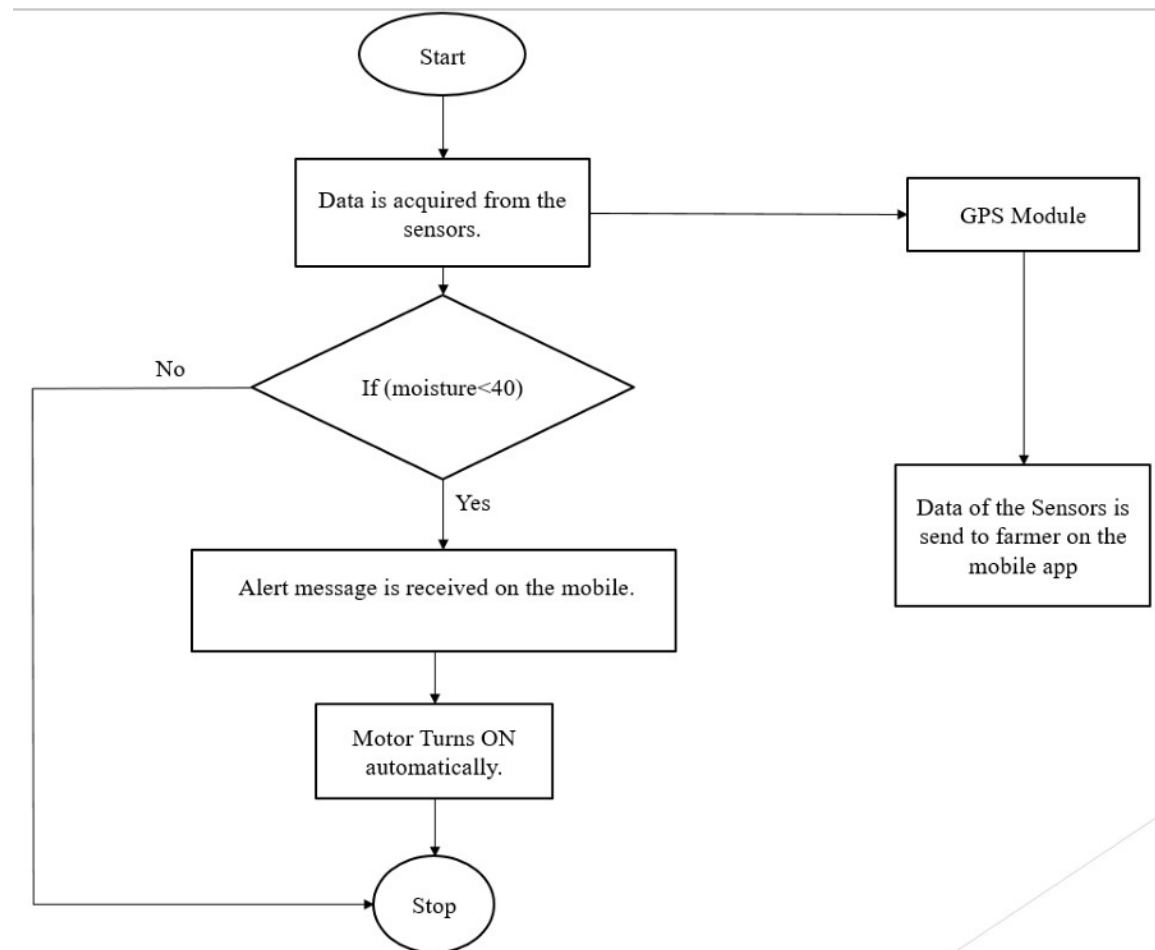


Figure 2.1 Block Diagram of Proposed Model

2.2 Flow Chart

When the data of different sensors that are humidity, temperature, soil moisture and location is acquired it is sent to the mobile app of the user and if the water content in the soil is less than the cut off value then an alert message is received on the app of the user and motor gets switched on automatically using relay.



2.3 Hardware Tools

2.3.1 NodeMCU

It is an open sources firmware and development kits to build IoT products. It includes firmware that run on ESP8266 WiFiSoC and hardware that has an ESP-12 module. The kit has analog (A0). It also has digital (D0-D8) pins on the board. It even assists serial ports communications such as SPI, UART, I2C etc.



Figure 2.2 NodeMCU

2.3.2 DHT11 Sensor

This sensor is basically a cost efficient digital humidity & temperature sensor. This sensor supplies digital output and therefore can be directly connected to data pins of microcontroller in spite of using ADC. It also consists of eight bit microcontroller to provide values of temperature & humidity in the form of data that is serial. It has 4 pins they are VCC, GND, DATA and NC. It operates from 3.3-5 volts power supply. This sensor has exceptional quality, anti-interference capability, economical performance and fast reaction benefits. Humidity is calculated by means of measuring the conductivity of liquid substrate that alters with exchange in humidity and temperature is calculated by the usage of a thermistor.

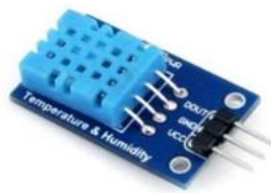


Figure 2.3 DHT11 Sensor

2.3.3 Soil Moisture Sensor

Moisture sensor has 3 pins – one is for voltage input, second for ground and third is for analog input. Moisture content of the soil (volume %) is measured by this sensor. The analog value need to be mapped in the range of 0-100 as moisture content is evaluated in percentage. The property used by this sensor is electrical resistance of soil. There are 2 probes in this sensor that permits the current to pass through the soil. After that it gets the value of resistance to measure the water content level. This implies that higher the water content higher is the conduction of electricity which means lesser resistance. If the soil is dry then the conduction in the soil is poor, this leads to increase in level of resistance. Hence it uses the property of resistance to measure the moisture in soil. It could be joined in two different ways they are Analog and Digital mode.

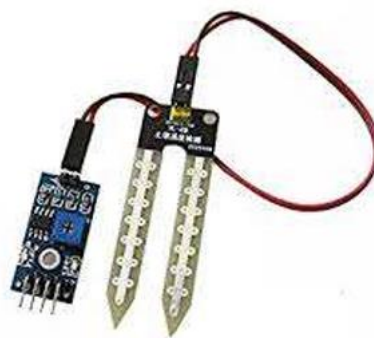


Figure 2.4 Soil Moisture Sensor

2.3.4 GPS Module

The full form of GPS is global positioning system so through this module anyone can always obtain the information regarding position anywhere in the world, generally used in smart phones. It is basically a satellite based system which uses satellite and ground stations to compute position on the earth. It needs to receive data from a minimum of 4 satellites for accuracy. So the NEO-6MV2 GPS module gives output in National Marine Electronics Association. This module provides data in the form of longitude and latitude positions.



Figure 2.5 GPS Module

2.3.5 Relay

It is a switching device. To mechanically control a switch many of the relays use electromagnet but some other fundamentals can also be used like relays that are solid state. When it is important to operate a circuit by a way of independent low power signal or if different circuits are managed by means of a single signal, then relays are used. So relay acts as an automated switch that operates on circuit having high current using low current signal.

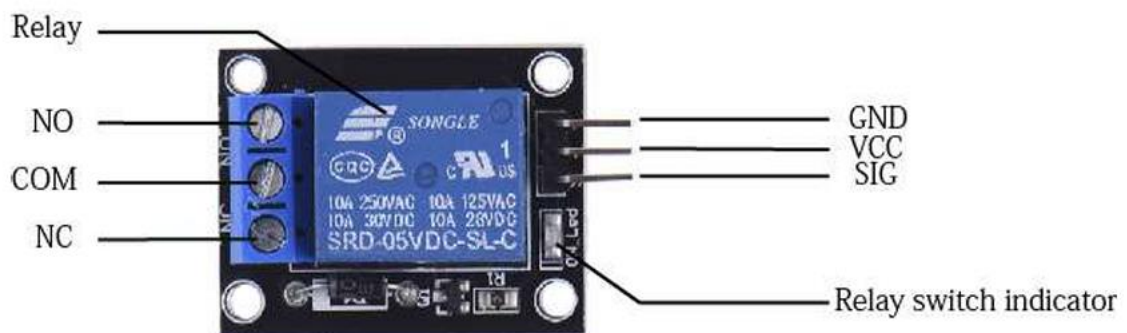


Figure 2.6 Pins of Relay

2.3.6 Motor

It is a micro submersible pump which works on dc 3-6v with cost efficient and portable. It is able to take around 120 litres for every hour with extremely low current utilization. Water level should be higher as if the motor is used without water it can harm the parts of this device due to overheating. There are many applications such as controlled fountain water flow, hydroponic systems, controlled garden watering system.



Figure 2.7: Motor

2.4 Software Tools

2.4.1 Arduino Ide (Integrated Development Environment)

Arduino IDE is an open source programming which is basically used to write & compile code using a module that is Arduino. This is an official programming software which makes compiling of code simple so a typical man can understand the learning procedure. This software is readily available for all operating systems like MAC, windows, Linux. Arduino Mega, Arduino Uno, Arduino Leonardo and more are range of Arduino modules that are available.

It basically has a text editor which is used for writing code, a text console, a message area, a toolbar with buttons for some of the common functions. Sketches are called as the programs that are written using this software. Coding on this software mostly uses functions of c/c++.



Figure 2.8 Arduino IDE

CHAPTER 3

RESULTS AND DISCUSSION

3.1 Circuit description

The Figure 3.1 shows the experimental setup of our smart agriculture monitoring system. The setup contains Microcontroller that is NodeMCU then sensors they are soil moisture, DHT11, GPS module and the last is Motor which is connected through relay. In this NodeMCU gives base for live streaming of temperature, humidity, soil moisture and sending the sensor information to the server using ESP8266 WiFi module and also the data of these sensors are send to the mobile app along with the GPS location. The sensors are interfaced with the microcontroller (NodeMCU) and are given power supply. Values from the sensors is read by NodeMCU and this microcontroller posts the information to the cloud server. When the value of moisture of the soil reaches below a certain threshold value which results the relay to get ON that leads to switching ON of the motor automatically and whenever the moisture value reaches the threshold level relay automatically switches OFF the motor.

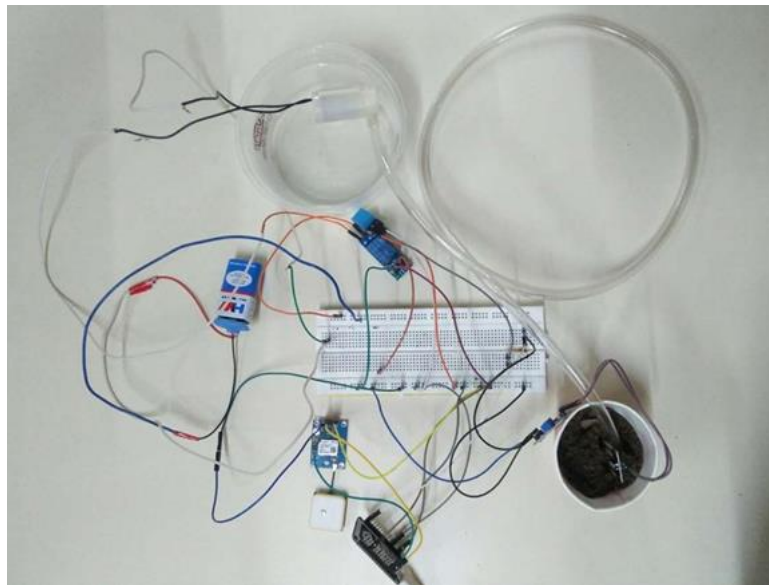


Figure 3.1 Hardware Setup

3.2 Results

The Smart Agriculture system was successfully implemented and tested to monitor environmental conditions and automate irrigation based on real-time sensor data. The system accurately collected information related to soil moisture, temperature, and humidity, enabling continuous observation of field conditions. The sensors provided

reliable readings, which were processed by the microcontroller to support automated decision-making. One of the most significant outcomes of the project was the efficient management of water resources. The automatic irrigation mechanism ensured that water was supplied only when the soil moisture level dropped below the predefined threshold. This reduced unnecessary water usage and prevented over-irrigation, resulting in more efficient water conservation compared to traditional irrigation methods.

The project also reduced the need for continuous manual monitoring of agricultural fields. Farmers can obtain real-time information about soil and environmental conditions through a connected dashboard or monitoring platform. This feature saves time and effort while improving the overall management of agricultural activities

Another important result was the improvement in crop health and productivity. Maintaining optimal soil moisture levels helps plants receive the required amount of water for growth. The continuous monitoring of temperature and humidity also provides valuable information that can be used to take preventive measures against unfavorable environmental conditions. The system demonstrated the advantages of integrating IoT technology into agriculture. Remote monitoring capabilities enabled users to access field data from any location with an internet connection. This increased convenience and allowed timely decisions regarding irrigation and crop management, thereby improving operational efficiency.

Overall, the Smart Agriculture system proved to be an effective and practical solution for modern farming. The project successfully combined sensing, automation, and communication technologies to create an intelligent agricultural system. The results indicate that IoT-based smart farming can enhance resource utilization, reduce labor requirements, improve crop management, and contribute to sustainable agricultural development.

CHAPTER 4

CONCLUSION AND FUTURE SCOPE

4.1 Conclusion

The proposed model explores the use of IoT (Internet of things) in the agriculture sector. This model aims at increasing the crop yield by helping in predicting better crop sequence for a particular soil. Thingspeak helps in real time sampling of the soil and hence the data acquired can be further used for analysing the crop. We have also taken many readings of the soil moisture, temperature and humidity of the environment for various days at different times of the day. Data on the cloud also helps the agriculturists in improving the yield, evaluating the manures, illness in the fields.

This system is cost effective and feasible. It also focuses on optimizing the use of water resources which combats issues like water scarcity and ensures sustainability. This model focuses on the utilization of IoT in agriculture and the solutions proposed in this paper will improve farming methods, increase productivity and lead to effective use of limited resources.

The Smart Agriculture project successfully demonstrated the application of Internet of Things (IoT) technology in modern farming. By integrating soil moisture, temperature, and humidity sensors with a microcontroller and automated irrigation system, the project enabled real-time monitoring and efficient management of agricultural resources. The system effectively reduced manual intervention by automatically controlling irrigation based on soil conditions.

The implementation of this project highlighted the benefits of IoT in improving water conservation, enhancing crop productivity, and supporting data-driven decision-making. Remote monitoring capabilities allowed users to access field information from anywhere, making farm management more convenient and efficient. The project also demonstrated how automation can help address challenges faced in traditional farming practices.

Overall, the Smart Agriculture system proved to be a reliable, cost-effective, and sustainable solution for smart farming. The successful completion of the project provided valuable insights into IoT technologies and their practical applications in agriculture. With further enhancements and the integration of advanced sensors and analytics, such systems have the potential to significantly contribute to the future of intelligent and sustainable agriculture.

4.2 Future Scope

The future scope of this project could be including variety of soil sensors like pH sensor, Rain sensor and then collecting and storing the data on cloud server. This would make the predicting and analysing processes more accurate. It also includes making different data mining algorithms suitable for data analysis in agriculture. The Smart Agriculture system can be further enhanced by integrating additional sensors such as pH sensors, rainfall sensors, nutrient sensors, and light intensity sensors. These sensors would provide more detailed information about soil and environmental conditions, enabling farmers to make better decisions regarding crop cultivation and resource management.

Future versions of the system can incorporate advanced technologies such as Artificial Intelligence (AI) and Machine Learning (ML) to analyze historical and real-time data. These technologies can help predict irrigation requirements, crop diseases, weather conditions, and yield outcomes, thereby improving agricultural productivity and efficiency.

The system can also be expanded by using cloud computing and big data analytics for large-scale agricultural applications. Farmers can store and analyze vast amounts of data collected from multiple fields, helping them identify trends and optimize farming practices. Mobile applications with user-friendly interfaces can further improve accessibility and ease of operation.

Overall, the future scope of Smart Agriculture is vast and promising. With continuous advancements in IoT, AI, and communication technologies, smart farming systems can play a significant role in increasing food production, conserving natural resources, reducing operational costs, and supporting sustainable agricultural development.



Figure 4.1: Rain Sensor



Figure 4.2: pH Sensor

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